EXP-2

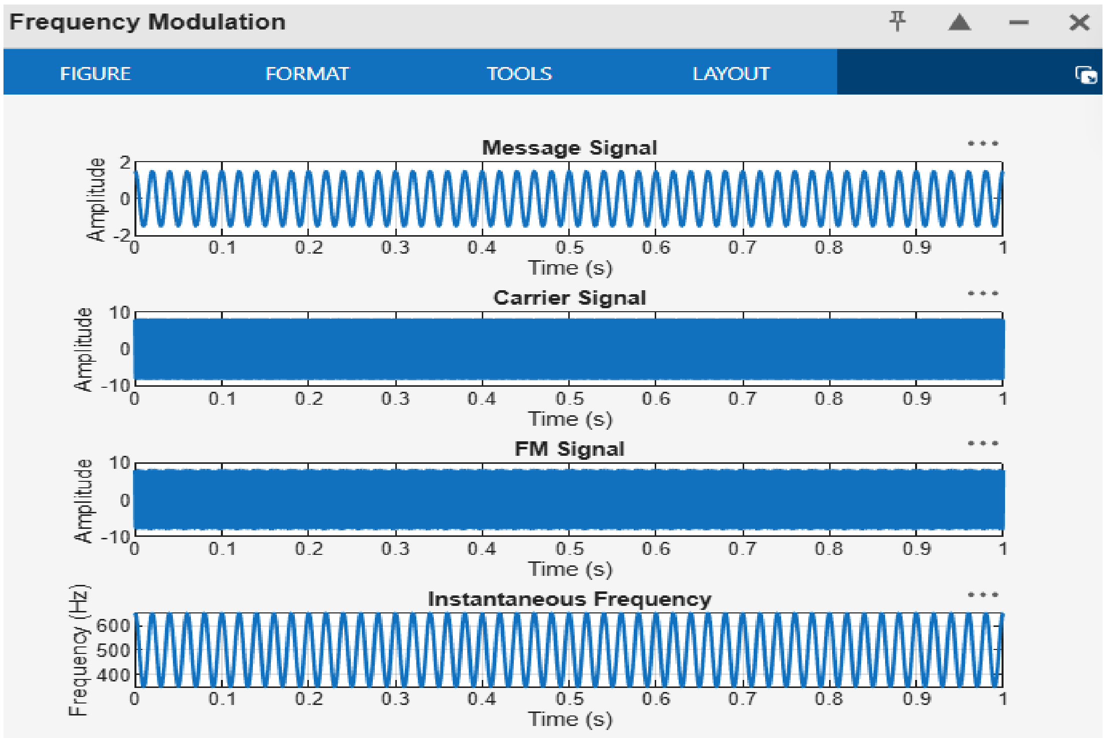
CODE:2A

```
clc;

clear;

close all;

Am = 1.5;

Ac = 8;

fm = 50;

fc = 500;

beta = 3;

fs = 10000;

t = 0:1/fs:1;

m = Am*cos(2*pi*fm*t);

c = Ac*cos(2*pi*fc*t);

fm_sig = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));

inst_freq = fc + beta*fm*cos(2*pi*fm*t);

figure('Name','Frequency Modulation','NumberTitle','off');

subplot(4,1,1);plot(t,m,'LineWidth',1.5);title('Message
Signal');xlabel('Time(s)');ylabel('Amplitude');grid on;

subplot(4,1,2);plot(t,c,'LineWidth',1.5);title('Carrier Signal');xlabel('Time
(s)');ylabel('Amplitude');grid on;

subplot(4,1,3);plot(t,fm_sig,'LineWidth',1.5);title('FM Signal');xlabel('Time
(s)');ylabel('Amplitude');grid on;

subplot(4,1,4);plot(t,inst_freq,'LineWidth',1.5);title('Instantaneous
Frequency');xlabel('Time (s)');ylabel('Frequency (Hz)');grid on;

fprintf('Message Frequency = %d Hz\n', fm);

fprintf('Carrier Frequency = %d Hz\n', fc);

fprintf('Modulation Index = %.2f\n', beta);
```

fprintf('Maximum Frequency Deviation = %.2f Hz\n', beta*fm);

fprintf('Bandwidth (Carson Rule) = %.2f Hz\n', 2*(beta*fm + fm));

OUTPUT :



CODE:2A.1

clc;

clear;

close all;

%% Input Parameters

Am = 1;   % Message signal amplitude (V)

fm = 1000;   % Message frequency (Hz)

Ac = 5;   % Carrier amplitude (V)

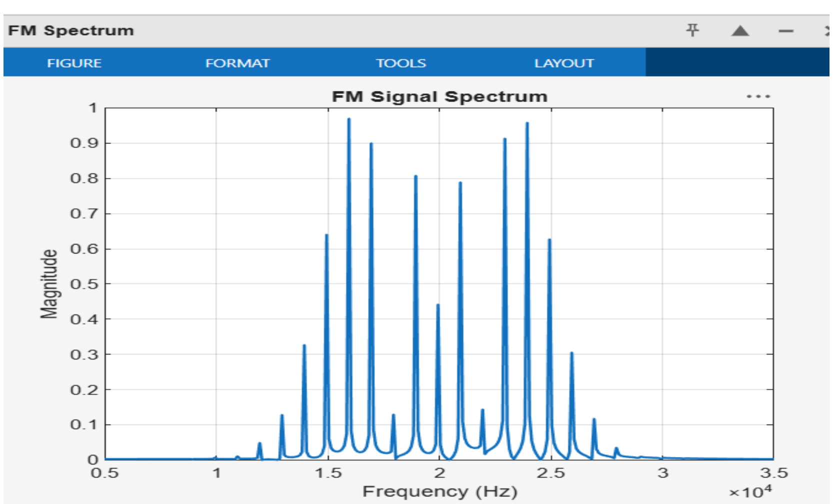
fc = 20000;   % Carrier frequency (Hz)

kf = 5000;   % Frequency sensitivity (Hz/V)

```
fs = 200000;     % Sampling frequency (Hz)

T = 0.01;        % Simulation duration (s)

t = 0:1/fs:T;

%% Message Signal

m = Am*cos(2*pi*fm*t);

%% Frequency Deviation

delta_f = kf * Am;

%% Modulation Index

beta = delta_f / fm;

%% FM Signal

fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));

%% Plot Message Signal

figure('Name','Frequency Modulation','NumberTitle','off');

subplot(2,1,1);plot(t,m,'LineWidth',1.5);title('MessageSigna');xlabel('Time
(s)');ylabel('Amplitude');grid on;

%% Plot FM Signal

subplot(2,1,2);plot(t,fm_signal,'LineWidth',1.5);title('Frequency Modulated
Signal');xlabel('Time (s)');ylabel('Amplitude');

grid on;

%% Spectrum Analysis

N = length(fm_signal);

FM_fft = fftshift(abs(fft(fm_signal))/N);

f = (-N/2:N/2-1)*(fs/N);

figure('Name','FM Spectrum','NumberTitle','off');

plot(f,FM_fft,'LineWidth',1.5);

title('FM Signal Spectrum');

xlabel('Frequency (Hz)');

ylabel('Magnitude');
```

```
grid on;
xlim([fc-15000 fc+15000]);
%% Display Results
fprintf('\nFM SIGNAL PARAMETERS\n');
fprintf('----------------------------\n');
fprintf('Carrier Frequency      = %.0f Hz\n', fc);
fprintf('Message Frequency      = %.0f Hz\n', fm);
fprintf('Message Amplitude      = %.2f V\n', Am);
fprintf('Carrier Amplitude      = %.2f V\n', Ac);
fprintf('Frequency Sensitivity  = %.0f Hz/V\n', kf);
fprintf('\nCALCULATED VALUES\n');
fprintf('----------------------------\n');
fprintf('Frequency Deviation (Delta f) = %.0f Hz\n', delta_f);
fprintf('Modulation Index (Beta)       = %.2f\n', beta);
%% Carson's Rule Bandwidth
BW = 2*(delta_f + fm);
fprintf('Bandwidth (Carson Rule) = %.0f Hz\n', BW);
```

OUTPUT :



CODE :2B

```
clc;
clear;
close all;
Am = 1;
fm = 1000;
Ac = 5;
fc = 20000;
%% FM Parameters
kf = 4000;
fs = 200000;
T = 0.01;
%% Time Vector
t = 0:1/fs:T;
%% Message Signal
m = Am*cos(2*pi*fm*t);
%% Frequency Deviation
delta_f = kf * Am;
%% Modulation Index
beta = delta_f / fm;
%% FM Signal
fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
%% Display Parameters
fprintf('Carrier Frequency (fc) = %.0f Hz\n', fc);
fprintf('Message Frequency (fm) = %.0f Hz\n', fm);
```

```
fprintf('Message Amplitude (Am) = %.2f V\n', Am);
fprintf('Carrier Amplitude (Ac) = %.2f V\n', Ac);
fprintf('Frequency Sensitivity (kf) = %.0f Hz/V\n', kf);
fprintf('\nFrequency Deviation (Delta f) = %.2f Hz\n', delta_f);
fprintf('Modulation Index (Beta) = %.2f\n', beta);
%% Verification
beta_verify = delta_f / fm;
fprintf('Verified Modulation Index = %.2f\n', beta_verify);
%% Verification Check
tolerance = 1e-6;
if abs(beta - beta_verify) < tolerance
    fprintf('\nVerification Successful\n');
else
    fprintf('\nVerification Failed\n');
end
%% Plotting
figure('Name','FM Modulation','NumberTitle','off');
subplot(2,1,1);
plot(t,m,'LineWidth',1.5);
grid on;
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
subplot(2,1,2);
plot(t,fm_signal,'LineWidth',1.2);
grid on;
title('FM Signal');
```

```
xlabel('Time (s)');
ylabel('Amplitude (V)');
```

OUTPUT :